

STATISTICAL TECHNIQUE

1) Correlation

(A) Spearman's rank

i) Non Repeated

$$R = 1 - \frac{6 \sum d^2}{n^3 - n}$$

ii) Repeated

$$R = 1 - \frac{6[\sum d^2 + \frac{1}{12}(m_1^3 - m_1) + \frac{1}{12}(m_2^3 - m_2) \dots]}{n^3 - n}$$

m = no. of times rank is repeated

$$d = R_1 - R_2 \quad \text{or} \quad R_2 - R_1$$

$$-1 \leq R \leq 1$$

(B) Karl Person's Coefficient of Correlation

$$i) \quad r = \frac{\sum dx dy - \frac{(\sum dx)(\sum dy)}{n}}{\sqrt{\sum d^2 x - \frac{(\sum dx)^2}{n}} \cdot \sqrt{\sum d^2 y - \frac{(\sum dy)^2}{n}}}$$

if x : , , , , [tabular form]
y : , , , ,

$$ii) \quad r = \frac{\sum xy - \frac{(\sum x)(\sum y)}{n}}{\sqrt{\sum x^2 - \frac{(\sum x)^2}{n}} \cdot \sqrt{\sum y^2 - \frac{(\sum y)^2}{n}}}$$

if $\sum x, \sum y, \sum xy$ is given

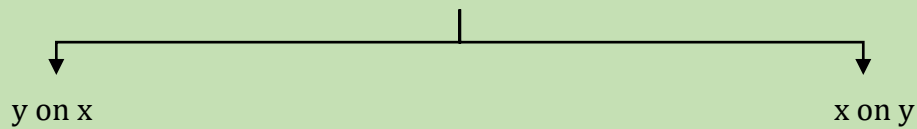
$$iii) \quad r = \frac{\frac{\sum (x - \bar{x})(y - \bar{y})}{n}}{\sqrt{\frac{\sum (x - \bar{x})^2}{n}} \cdot \sqrt{\frac{\sum (y - \bar{y})^2}{n}}}$$

if $\sum (x - \bar{x}), \sum (y - \bar{y}),$
 $\sum (x - \bar{x})(y - \bar{y})$ is given

2) Regression

(A) Regression Coefficient:

Regression Coefficient (b)



$$\text{i) } b_{yx} = r \frac{\sigma_y}{\sigma_x}$$

$$b_{xy} = r \frac{\sigma_x}{\sigma_y}$$

$$\text{ii) } = \frac{\sum dx dy - \frac{(\sum dx)(\sum dy)}{n}}{\sum dx^2 - \frac{(\sum dx)^2}{n}}$$

$$= \frac{\sum dx dy - \frac{(\sum dx)(\sum dy)}{n}}{\sum dy^2 - \frac{(\sum dy)^2}{n}}$$

$$\text{iii) } = \frac{\sum xy - \frac{(\sum x)(\sum y)}{n}}{\sum x^2 - \frac{(\sum x)^2}{n}}$$

$$= \frac{\sum xy - \frac{(\sum x)(\sum y)}{n}}{\sum y^2 - \frac{(\sum y)^2}{n}}$$

(B) Lines of Regression

i) y on x

$$y - \bar{y} = b_{yx}(x - \bar{x})$$

ii) x on y

$$x - \bar{x} = b_{xy}(y - \bar{y})$$

(C) Angle between Lines

$$\tan \theta = \left(\frac{1 - r^2}{r} \right) \left(\frac{\sigma_x \cdot \sigma_y}{\sigma x^2 + \sigma y^2} \right)$$

(D) Note:

$$\text{(i) } r = \sqrt{(b_{xy})(b_{yx})}$$

(ii) Sign of 'r' is same as sign of 'b_{xy}' & 'b_{yx}'

3) Curve Fitting

(A) Straight Line

$$\text{i) } y = a + bx$$

$$\sum 1 = n$$

$$\sum y = \sum a + \sum bx$$

$$\sum y = a \sum 1 + b \sum x$$

$$\sum y = an + b \sum x$$

ii) $yx = ax + bx^2$

$$\sum yx = \sum ax + \sum bx^2$$

$$\sum yx = a \sum x + b \sum x^2$$

(B) Parabola

i) $y = a + bx + cx^2$

$$\sum y = \sum a + \sum bx + \sum cx^2$$

$$\sum y = an + b \sum x + c \sum x^2$$

ii) $yx = ax + bx^2 + cx^3$

$$\sum xy = a \sum x + b \sum x^2 + c \sum x^3$$

iii) $x^2y = ax^2 + bx^3 + cx^4$

$$\sum x^2y = a \sum x^2 + b \sum x^3 + c \sum x^4$$